Problem Set 5 Solution

Part One: Computer exercises

1. Download the data set `china` from https://www.openintro.org/data/, which was collected from the 2006 China Health and Nutrition Survey conducted by the UNC Carolina Population Center. The Survey aims to examine the effects of the health, nutrition, and family planning policies and programs implemented by national and local governments. The variable definition is summarized in the following table.

   For the analysis below, you need to first drop the observations with a missing value for any of the variables.

```
# clean data: drop samples with unknown education degree, or unknown child care hours
china_clean <- china %>%
   filter(edu != 9 & is.na(edu) == FALSE) %>%
   filter(child_care != -99 & is.na(child_care) == FALSE)
```

   1. Make a summary statistics table that includes the number of observations, mean, standard deviation, minimum, and maximum for `child_care`. Make a histogram for `child_care`.

```
# summary statistics table
china_clean %>% summarize(
    count = n(),
    mean = mean(child_care),
    sd = sd(child_care),
    min = min(child_care),
    max = max(child_care)
  )

# output
  count     mean      sd min max
1   799 25.01001 28.6975   1 168

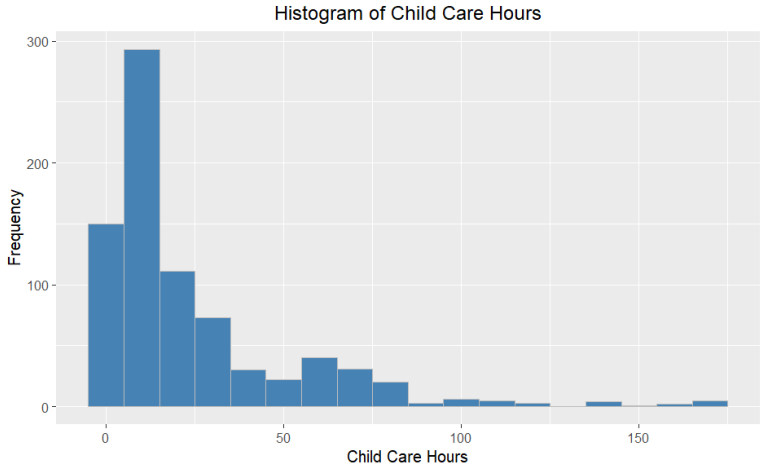
# plot a histogram for `child_care`
ggplot(china_clean, aes(x = child_care)) +

  ## select an appropriate bin width
  geom_histogram(binwidth = 10, fill = "steelblue", color = "gray") +

  ## add label and graph title to make the graph more readable
```

```
  labs(
    title = "Histogram of Child Care Hours",
    x = "Child Care Hours",
    y = "Frequency"
  ) +

  ## center the graph title
  theme(plot.title = element_text(hjust = 0.5))
```



2. Make a summary statistics table that includes the number of observations, mean, standard deviation, minimum, and maximum for `child_care` by parents' `gender`. Make histograms of `child_care` by parents' `gender`. Compare the distribution of `child_care` between males and females.

```
# summary statistics table: by parent's gender
china_clean %>%
  group_by(gender) %>%
  summarize(
    count = n(),
    mean = mean(child_care),
    sd = sd(child_care),
    min = min(child_care),
    max = max(child_care)
  )

# output
  gender count  mean    sd   min   max
   <int> <int> <dbl> <dbl> <int> <int>
1      1   312  15.7  20.7     1   168
2      2   487  31.0  31.4     1   168
```

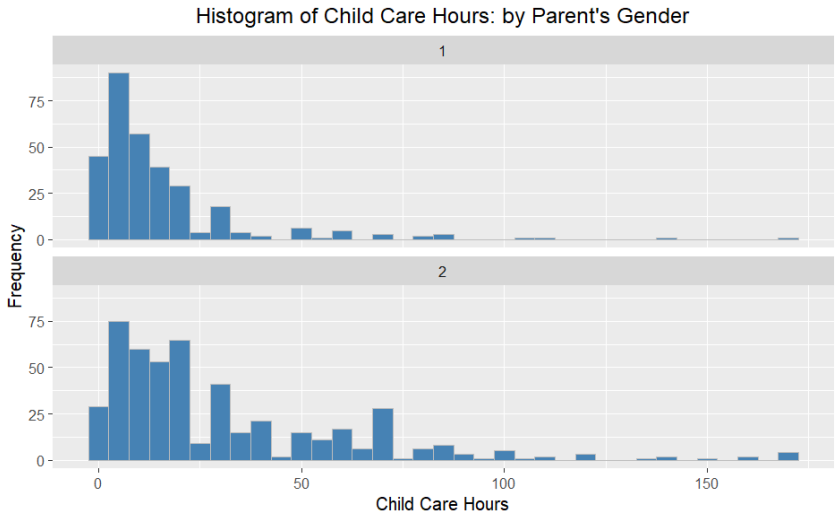
```r
# plot a histogram for `child_care`: by parent's gender
ggplot(china_clean, aes(x = child_care)) +

  ## select an appropriate bin width
  geom_histogram(binwidth = 5, color = "gray", fill = "steelblue") +

  ## split the sample by parent's gender, set the nrow = 2 for comparis
on
  facet_wrap(~gender, nrow = 2) +

  ## add label and graph title to make the graph more readable
  labs(
    title = "Histogram of Child Care Hours: by Parent's Gender",
    x = "Child Care Hours",
    y = "Frequency",
    fill = "Gender"
  ) +

  ## center the graph title
  theme(plot.title = element_text(hjust = 0.5))
```



3. Is there a difference in the average childcare hours between males and females? Draw a conclusion using both the critical value approach and p-value approach.

```r
# t-test
t_test_result <- t.test(child_care ~ gender, data = china_clean)
print(t_test_result)
```

```
# output
    Welch Two Sample t-test

data:  child_care by gender
t = -8.2549, df = 796.45, p-value = 6.291e-16
alternative hypothesis: true difference in means between group 1 and gr
oup 2 is not equal to 0
95 percent confidence interval:
 -18.83770 -11.59989
sample estimates:
mean in group 1 mean in group 2
       15.73397        30.95277
```

Set $\alpha = 0.05$. The absolute value of the t-statistic (-8.25) is greater than the critical value of 1.96. Besides, the p-value is less than 0.05. So we reject H_0, implying the average childcare hours is significantly different between males and females.

4. Make a summary statistics table that includes the number of observations, mean, standard deviation, minimum, and maximum for `child_care` by parents' education levels. Make side-by-side box plots for `child_care` by parents' education levels. Compare the distribution of child_care among parents' education levels.

```
# summary statistics table: by parent's education level
china_clean %>%
    group_by(edu) %>%
    summarise(
      count = n(),
      mean = mean(child_care),
      sd = sd(child_care),
      min = min(child_care),
      max = max(child_care)
    )

# output
    edu count  mean    sd   min   max
  <int> <int> <dbl> <dbl> <int> <int>
1     1   223  25.4  30.0     1   168
2     2   351  24.7  27.6     1   168
3     3    80  22.2  28.8     1   161
4     4    78  30.9  33.4     1   140
5     5    67  21.8  23.3     1   108

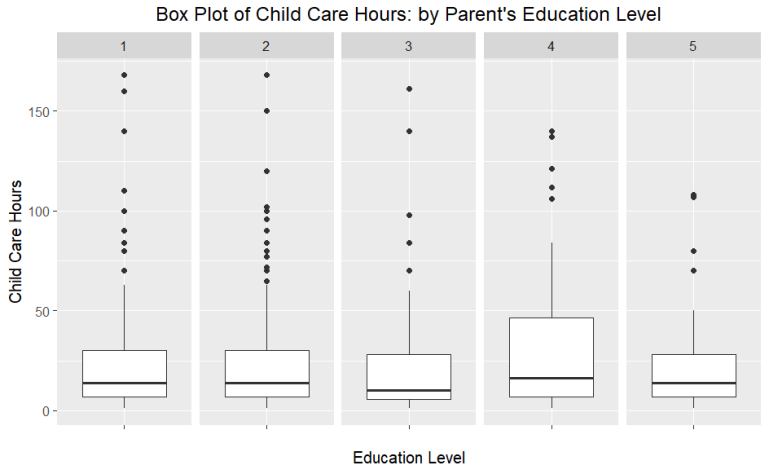
# plot side-by-side box plots for `child_care`: by parent's educational
level
ggplot(china_clean, aes(y = child_care)) +
```

```
  ## remove the scale of x-axis
  geom_boxplot(aes(x = "")) +

  ## split the sample by education level, set nrow = 1 for comparison
  facet_wrap(~edu, nrow = 1) +

  ## add label and graph title to make the graph more readable
  labs(
    title = "Box Plot of Child Care Hours: by Parent's Education Leve
l",
    x = "Education Level",
    y = "Child Care Hours"
  ) +

  ## center the graph title
  theme(plot.title = element_text(hjust = 0.5))
```



5. Do average childcare hours vary with parents' education? Draw a conclusion using both the critical value approach and p-value approach.

```
# summary statistics table: by parent's education level
anova_result <- aov(child_care ~ as.factor(edu), data = china_clean)
print(anova_result)
summary(anova_result)

# output
Call:
   aov(formula = child_care ~ as.factor(edu), data = china_clean)
```

```
Terms:
                as.factor(edu) Residuals
Sum of Squares          4142.1  653047.8
Deg. of Freedom              4       794

Residual standard error: 28.67888
Estimated effects may be unbalanced
                Df Sum Sq Mean Sq F value Pr(>F)
as.factor(edu)   4   4142  1035.5   1.259  0.285
Residuals      794 653048   822.5
```

Set $\alpha = 0.05$. The F-statistics (1.259) is less than the critical value of X. Besides, the p-value (0.285) is greater than 0.05. Thus, we do not reject H_0. We don't have strong evidence that the average childcare hours vary with parents' education.

Part Two: Textbook exercises

Question 7.4

a. We have $n = 26,\ T = 2.485,\ df = 26 - 1 = 25$, and p-value $= 0.020$. So we do not reject H_0

b. We have $n = 18,\ T = 0.5,\ df = 18 - 1 = 17$, and p-value $= 0.623$. So we do not reject H_0

Question 7.12

a.

$$H_0 : \mu = 35, \quad H_A : \mu \neq 35$$

b.

- **Independence:** if we can assume that these 52 officers represent a random sample (big assumption), then independence would be satisfied, but we cannot check this.
- **Normality:** we don't have a plot of the distribution that we can use to check this condition. We at least have more than 30 observations, so the distribution would have to be extremely skewed to be an issue. We again cannot check this, but this seems like a less concerning issue than the independence consideration.

c. The test statistic and the p-value can be calculated as follows:

$$T = \frac{124.32 - 35}{\frac{37.74}{\sqrt{52}}} \approx 17.07$$

$$df = 52 - 1 = 51$$

$$\text{p-value} = 2 \times P(T_{51} > 17.07) < 0.001$$

The hypothesis test yields a very small p-value, so we **reject** H_0. Given the direction of the data, there is very convincing evidence that the police officers have been exposed to a higher concentration of lead than individuals living in a suburban area.

Question 7.20

a. The median writing score is slightly higher but it's difficult to tell if the average scores on the two tests are different or not.

b. No, the score of one student on the reading test is **not** independent of their score on the writing test.

c. Let diff = read - write. Then the hypotheses are:

$$H_0 : \mu_{\text{diff}} = 0, \quad H_A : \mu_{\text{diff}} \neq 0$$

d. The conditions for the sampling distribution of $\bar{x}_{\text{diff}}$ to be nearly normal and the estimate of the standard error to be sufficiently accurate are as follows:

- **Independence:** Students are randomly sampled and $200 < 10\%$ of all students who take this survey, therefore we can assume that the reading and writing scores of one student are independent of another.
- **Normality:** The distribution of the differences appears fairly symmetric, so we can assume that the sampling distribution of average differences will be approximately normal.

e. The test statistic and the p-value can be calculated as follows:

$$T = \frac{\bar{x}_{\text{diff}} - \mu_{\text{diff}}}{s_{\text{diff}}/\sqrt{n}} = \frac{-0.545 - 0}{8.887/\sqrt{200}} \approx -0.87$$

$$df = 200 - 1 = 199$$

$$\text{p-value} = P(|T_{199}| > 0.87) > 0.2$$

Since the p-value > 0.05, **fail to reject** H_0. The data do not provide convincing evidence of a difference between the average reading and writing scores.

f. We may have made a **Type 2 error**, i.e. we may have incorrectly failed to reject H_0. In this context, a Type 2 error means deciding that the data do not provide convincing evidence of a difference between the average reading and writing scores of students when in reality there **is** a difference.

g. Since we failed to reject H_0, which claimed the average difference is equal to 0, we would expect a confidence interval to include this value.

Question 7.28

The hypotheses are as follows:

$$H_0 : \mu_{A,c} = \mu_{M,c}, \quad H_A : \mu_{A,c} \neq \mu_{M,c}$$

We are told to assume that conditions for inference are satisfied. Then, the test statistic and the p-value can be calculated as follows:

$$\begin{aligned} T &= \frac{(\bar{x}_{A,c} - \bar{x}_{M,c}) - (\mu_{A,c} - \mu_{M,c})}{\sqrt{\frac{s_{A,c}^2}{n_{A,c}} + \frac{s_{M,c}^2}{n_{M,c}}}} \\ &= \frac{(16.12 - 19.85) - 0}{\sqrt{\frac{3.58^2}{26} + \frac{4.51^2}{26}}} \\ &= \frac{-3.37}{1.13} \\ &\approx -3.3 \end{aligned}$$

For computational convenience,

$$df = min(n_{M,c} - 1, n_{A,c} - 1) = 25$$

$$\text{p-value} = P(|T_{25}| > 3.3) < 0.01$$

Since p-value < 0.05, reject H_0. The data provide strong evidence that there is a difference in the average city mileage between cars with automatic and manual transmissions.

Question 7.30

Follow the results in Question 7.28, we know $df = 25$ and thus we obtain $t_{25}^* = 2.49$

$$\begin{aligned} &(\bar{x}_{A,hwy} - \bar{x}_{M,hwy}) \pm t_{df}^* \sqrt{\frac{s_{A,hwy}^2}{n_{A,hwy}} + \frac{s_{M,hwy}^2}{n_{M,hwy}}} \\ =&(22.92 - 27.88) \pm 2.49 \times \sqrt{\frac{5.29^2}{26} + \frac{5.01^2}{26}} \\ =&-4.96 \pm 3.56 \\ =&(-8.52, -1.4) \end{aligned}$$

We are 98% confident that on the highway cars with **manual** transmissions get on average **1.4 to 8.52 MPG more** than cars with **automatic** transmissions.

Question 7.34

Since $\alpha = 0.05$ and we are using a two-sided test

$$Z_{\alpha/2} = 1.96$$

For power of 80%, $Z_\beta = 0.84$

Hence,

$$0.5 = 0.84 \times SE + 1.96 \times SE$$

$$SE = 0.1786 = \sqrt{\frac{(SD)^2}{n} + \frac{(SD)^2}{n}} = \sqrt{\frac{(2.2)^2}{n} + \frac{(2.2)^2}{n}}$$

Eventually, we attain $n = 303.47$. That is, we need 304 new enrollees

Question 7.40

The conditions that need to be satisfied for ANOVA are:

1. **Independence:** We are not told if the students are randomly assigned to discussion sections, so we cannot be sure of independence of observations. That is, these data will not allow us to attribute any differences in scores to the teaching assistants since these data are observational. Students with different academic strengths may have tended to enroll in different sections.
2. **Approximately normal:** We are not given plots of the distributions of grades by discussion section, therefore we cannot check this condition. Also, the sample sizes are not all large, so we can't relax this condition.
3. **Constant variance:** Based on the standard deviations given in the summary table, the constant variance assumption appears to be reasonable for most groups, but not all. Also, the sample sizes aren't consistent so we can't relax this condition.

In order to proceed with the test, we will need to assume independence and approximately normal distributions within groups.

The hypotheses are as follows:

$$H_0 : \mu_1 = \mu_2 = ... = \mu_8, \quad H_A : \text{The average score varies across some groups}$$

With a p-value > 0.05, we fail to reject H_0. The data do not provide convincing evidence that the average score varies across some (or all) groups.